

Michaela Vance

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website: <https://michaelavance.github.io/michaelavance/>

Education

- **University of Oklahoma Health Sciences Center, Oklahoma City, OK** August 2022-Present
GPA- 3.955
Classification- Doctoral Candidate
- **University of Central Oklahoma, Edmond, OK** May 2022
Bachelor of Science in Biology-Biomedical Sciences with Honors
GPA- 4.0
Summa Cum Laude
Classification- Alumni
- **Harrah High School, Harrah, OK** May 2018
High School Diploma
GPA- 4.25
Classification- Alumni

Research Experience

- **University of Oklahoma Health Sciences Center, Oklahoma City, OK** August 2022-Present
Graduate Research Assistant, Department of Cell Biology
Advisor- Shannon M. Conley, Ph.D.
 - Theoretical and experimental analysis of the role of cellular senescence and endothelial-to-mesenchymal transition during cerebrovascular aging.**Laboratory Skills:**
 - tissue embedding and staining, mammalian cell culture, DNA and RNA isolation, PCR, gel electrophoresis, phase contrast and fluorescent microscopy, mouse husbandry and neurological assessment, mouse surgical techniques, generation and analysis of biological data including bioinformatic analysis of bulk and single-cell RNAseq datasets.
- **University of Central Oklahoma, Edmond, OK** June 2019-May 2022
Undergraduate Research Assistant, Department of Biology
Advisor- Nikki J. Seagraves, Ph.D.
 - Theoretical and experimental analysis of the effects of phenylalanine, retinoic acid, retinal, and citral exposure on the proliferation of O9-1 mouse cranial neural crest cells**Laboratory Skills:**
 - tissue embedding and staining, chick embryo dissection, immunohistochemistry, mammalian and bacterial cell culture, DNA and RNA isolation, PCR, gel electrophoresis, cloning, phase contrast and fluorescent microscopy, research protocol optimization, generation and analysis of biological data

Employment

- **University of Oklahoma Health Sciences Center, Oklahoma City, OK** August 2022-Present
Graduate Research Assistant, Department of Cell Biology
Advisor- Shannon M. Conley
- **University of Central Oklahoma, Edmond, OK** January 2021-May 2022
Cell Biology Supplemental Instructor, College of Math and Science
Advisor- Nikki J. Seagraves, Ph.D.
- **University of Central Oklahoma, Edmond, OK** June 2019-Present
Undergraduate Research Assistant, Department of Biology
Advisor- Nikki J. Seagraves, Ph.D.

Leadership

- **Crimson Club, University of Oklahoma Health Sciences Center** August 2024-Present
 - **Member-** (August 2024-Present) members of Crimson Club must be nominated and go through an application process, once appointed they serve as ambassadors for the university conducting potential student tours and engaging in volunteer and outreach work
- **Academic Appeals Board, University of Oklahoma Health Sciences Center** August 2024-Present
 - **Member-** (August 2024-Present) member of the pool of potential graduate students that can serve on the panel for Graduate College appeals and academic misconduct cases
- **Student Government Association, University of Oklahoma Health Sciences Center** August 2023-May 2024
August 2025-Present
 - **Executive Council Member-** (August 2025-Present) attend meetings, communicate Graduate College events and graduate student needs to the larger student body
 - **Senator-** (August 2023-May 2024) attended meetings, worked to improve student experience on campus, represented graduate student interests to the larger student body
- **Graduate Student Association, University of Oklahoma Health Sciences Center** August 2022-Present
 - **President-** (August 2025-Present) attend and preside over meetings, serve as a representative on the Graduate Faculty Council and the student government's executive council, represent the graduate student body
 - **Vice President-** (August 2024-May 2025) attended and took minutes for meetings, distributed meeting minutes, and assisted the president with duties
 - **Senator-** (August 2023-May 2024) attended meetings, sat as a senator on the student government, relayed graduate student needs to the larger student body, updated graduate students on campus events
 - **Member-** (August 2022-May 2023) attended meetings and social events
- **Tri-Beta Biological Honor Society, University of Central Oklahoma** August 2019-May 2022
 - **President-** (January 2021-May 2022) organized events, invited guest speakers, planned meetings, planned and conducted induction, recruited members

- **Vice President-** (August 2020-December 2021) organized events, planned meetings, planned and conducted induction, recruited members
- **Sergeant of Arms-** (August 2019-May 2020) organized events, planned meetings, planned and conducted induction, recruited members

Professional Organizations

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|-------------------------------------|--------------|
| • American Heart Association | 2024-2025 |
| - Member | |
| • American Aging Association | 2022-Present |
| - Member | |
| • Society for Developmental Biology | 2020-2022 |
| - Member | |

Honors and Awards

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| • Graduate Student Data Blitz Award Winner American Aging Association 52 nd Annual Meeting | May 2025 |
| • Travel Award, College of Medicine's Dean's Award For Scientific Achievement University of Oklahoma Health Sciences Center Graduate Research Education and Technology (GREAT) Symposium | April 2025 |
| • Travel Award, Best Graduate Student Poster Presentation University of Oklahoma Health Sciences Center Cell Biology Symposium | October 2024 |
| • College of Mathematics and Science Department of Biology Outstanding Senior Award | May 2022 |
| • College of Mathematics and Science Department of Biology Academic Achievement Award | May 2022 |
| • Dr. Ethel Derrick Endowed Scholarship for Biology | August 2021 |
| • Society for Developmental Biology 80 th Annual Meeting Best Student Poster Finalist- Top 6 | Summer 2021 |
| • OUHSC Summer Undergraduate Research Program Acceptance | Summer 2020, Summer 2021 |
| - Summer 2020 Program Cancelled Due to COVID-19 | |
| • College of Mathematics and Science Department of Chemistry Organic Chemistry Award for Outstanding Performance in Organic Chemistry | Spring 2021 |
| • Beta Beta Beta Frank G. Brooks Award for Outstanding Oral Presentation- 2 nd Place | Spring 2021 |
| • UCO President's Honor Roll | Fall 2018, Spring 2019
Fall 2019, Spring 2020,
Fall 2020, Spring 2021,
Fall 2021, Spring 2022 |
| • Harrah High School Valedictorian | May 2018 |
| • UCO Freshman Achievement Waiver | August 2018 |
| • Gear Up Scholarship | August 2018 |

Funding

- **Student Research, Creative, and Scholarly Activities (RCSA) Grant-** August 2021-May 2022
Assessing the Effect of Phenylalanine on Neural Crest Cells In Vivo (001298)
- **OUHSC Summer Undergraduate Research Programs: The Oklahoma IDeA Network of Biomedical Research Excellence Grant-** Summer 2021
Effect of Phenylalanine on Proliferation and Apoptosis of O9-1 Mouse Cranial Neural Crest Cells
- **Student Research, Creative, and Scholarly Activities (RCSA) Grant-** August 2020-May 2021
Assessing the Effect of Phenylalanine, Retinoic Acid, and Retinoic Acid Inhibitor on the Migration of O9-1 Mouse Cranial Neural Crest Cells (001075)
- **The Oklahoma IDeA Network of Biomedical Research Excellence Student Mentoring and Research Training SMART-** Summer 2019
Assessing the Effect of Phenylalanine, Retinoic Acid, and Diethylaminobenzaldehyde on Myocardial Cells.

Publications

- Pinckard J*, Negri S*, Huston CA*, Bickel MA, **Vance ML**, Milan M, Hibbs CL, Budda M, Chandragiri SS, Pipkin K, Tarantini S, Conley SM. Functional ultrasound as a quantitative approach for measuring functional hyperemia in aging models. *Neuroimage*. 2025 Aug 1;316:121313. doi: 10.1016/j.neuroimage.2025.121313. Epub 2025 Jun 12. PMID: 40513179; PMCID: PMC12257925.
- Huston CA, Milan M, **Vance ML**, Bickel MA, Miller LR, Negri S, Hibbs C, Vaden H, Hayes L, Csiszar A, Ungvari Z, Yabluchanskiy A, Tarantini S, Conley SM. The effects of time restricted feeding on age-related changes in the mouse retina. *Exp Gerontol*. 2024 Sep;194:112510. doi: 10.1016/j.exger.2024.112510. Epub 2024 Jul 5. PMID: 38964431; PMCID: PMC11425985.
- Miller LR, Bickel MA, **Vance ML**, Vaden H, Nagykaladi D, Nyul-Toth A, Bullen EC, Gautam T, Tarantini S, Yabluchanskiy A, Kiss T, Ungvari Z, Conley SM. Vascular smooth muscle cell-specific Igf1r deficiency exacerbates the development of hypertension-induced cerebral microhemorrhages and gait defects. *Geroscience*. 2024 Jun;46(3):3481-3501. doi: 10.1007/s11357-024-01090-7. Epub 2024 Feb 23. PMID: 38388918; PMCID: PMC11009188.
- Miller LR*, Bickel MA*, Tarantini S, Runion ME, Matakchiera Z, **Vance ML**, Hibbs C, Vaden H, Nagykaladi D, Martin T, Bullen EC, Pinckard J, Kiss T, Howard EW, Yabluchanskiy A, Conley SM. IGF1R deficiency in vascular smooth muscle cells impairs myogenic autoregulation and cognition in mice. *Front Aging Neurosci*. 2024 Feb 15;16:1320808. doi: 10.3389/fnagi.2024.1320808. PMID: 38425784; PMCID: PMC10902040.
- Bickel MA*, Sherry DM*, Bullen EC, **Vance ML**, Jones KL, Howard EW, Conley SM. Microvascular smooth muscle cells exhibit divergent phenotypic switching responses to platelet-derived growth factor and insulin-like growth factor 1. *Microvasc Res*. 2024 Jan;151:104609. doi: 10.1016/j.mvr.2023.104609. Epub 2023 Sep 15. PMID: 37716411; PMCID: PMC10842624.

Poster Presentations

- "Senescence and Vascular Transdifferentiation in the Aging Cerebral Vasculature." *American Aging Association 53rd Annual Meeting* (2025 May, Anchorage, Alaska)

- “Senescence and Endothelial to Mesenchymal Transition in the Aging Cerebral Vasculature.” *Graduate Research Education and Technology (GREAT) Symposium* (2025 April, Oklahoma City, Oklahoma)
 - **Travel Award, College of Medicine’s Dean’s Award for Scientific Achievement**
- “The Role of Age and IGF-1 Deficiency in Transdifferentiation of Cells in the Vascular Unit.” *Oklahoma Geroscience Symposium* (2025 April, Oklahoma City, Oklahoma)
- “The Role of Age and IGF-1 Deficiency in Transdifferentiation of Cells in the Vascular Unit.” *IGF and Insulin System in Physiology and Disease Gordon Research Conference* (2025 March, Ventura, California)
- “The Role of Senescence and Endothelial to Mesenchymal Transition in the Aging Cerebral Vasculature.” *University of Oklahoma Health Sciences Center Cell Biology Symposium* (2024 October, Oklahoma City, Oklahoma)
 - **Travel Award, Best Graduate Student Poster Presentation**
- “Senescence and Endothelial to Mesenchymal Transition in the Aging Cerebral Vasculature.” *American Aging Association 52nd Annual Meeting* (2024 June, Madison, Wisconsin)
- “The Role of Senescence and Endothelial to Mesenchymal Transition in the Aging Cerebral Vasculature.” *Web of Life Conference* (2024 April, Oklahoma City, Oklahoma)
- “The Role of Senescence and Endothelial to Mesenchymal Transition in the Aging Cerebral Vasculature.” *Oklahoma Geroscience Symposium* (2024 April, Oklahoma City, Oklahoma)
- “The Role of Senescence and Endothelial to Mesenchymal Transition in the Aging Cerebral Vasculature.” *Graduate Research Education and Technology (GREAT) Symposium* (2024 April, Oklahoma City, Oklahoma)
- “The Role of IGF-1 on Vascular Smooth Muscle Cells in the Development of Cerebral Microhemorrhages.” *American Aging Association 51st Annual Meeting* (2023 June, Oklahoma City, Oklahoma)
- “Cloning of Alcohol Dehydrogenase 1C for Enzyme Kinetics.” *Society for Developmental Biology 81st Annual Meeting* (July 2022, Vancouver, BC, Canada)
- “Cloning of Alcohol Dehydrogenase 1C for Enzyme Kinetics.” *The University of Central Oklahoma College of Mathematics and Science Tri-Center Symposium* (April 2022, Edmond, OK)
- “Cloning of Alcohol Dehydrogenase 1C for Enzyme Kinetics.” *Oklahoma Research Day* (March 2022, Virtual)
- “Effect of Phenylalanine, Retinoic Acid, and Citral on Proliferation and Apoptosis of O9-1 Mouse Cranial Neural Crest Cells.” *OUHSC Summer Undergraduate Research Programs Conference* (July 2021, Oklahoma City, OK)
- “Effect of Phenylalanine, Retinoic Acid, and Citral on Proliferation O9-1 Mouse Cranial Neural Crest Cells.” *Society for Developmental Biology 80th Annual Meeting* (July 2021, Virtual)
 - **Best Student Poster Finalist (top 6)**
- “Effect of Phenylalanine, Retinoic Acid, and Citral on Proliferation O9-1 Mouse Cranial Neural Crest Cells.” *Council on Undergraduate Research Posters on the Hill* (April 2021, Virtual)
- “Effect of Phenylalanine, Retinoic Acid, Retinal, and Citral on Proliferation of O9-1 Mouse Cranial Neural Crest Cells.” *Society for Developmental Biology 79th Annual Meeting* (July 2020, Virtual)
- “Effect of Phenylalanine, Retinoic acid, and 4-Diethylaminobenzaldehyde on Proliferation of O9-1 Mouse Cranial Neural Crest Cells.” *Oklahoma Academy of Science Technical Meeting* (November 2019, Edmond, OK)

Oral Presentations

- “Senescence and Vascular Transdifferentiation in the Aging Brain Microvasculature.” *American Aging Association 53rd Annual Meeting* (May 2025, Anchorage, Alaska)
 - Flash talk, **Graduate Student Data Blitz Award Winner**
- “Senescence and Endothelial to Mesenchymal Transition in the Aging Cerebral Vasculature.” *University of Oklahoma Health Sciences Center Cell Biology Department Seminar* (November 2024, Oklahoma City, OK)
- “Effect of Phenylalanine, Retinoic Acid, and Citral on Proliferation O9-1 Mouse Cranial Neural Crest Cells.” *Beta Beta Beta South-Central Regional Convention* (April 2021, Virtual)
 - **Beta Beta Beta Frank G. Brooks Award for Outstanding Oral Presentation- 2nd Place**
- “Effect of Phenylalanine, Retinoic Acid, and Citral on Proliferation O9-1 Mouse Cranial Neural Crest Cells.” *Oklahoma Research Day* (March 2021, Virtual)
- “Effect of Phenylalanine, Retinoic acid, Retinal, and Citral on Proliferation of O9-1 Mouse Cranial Neural Crest Cells.” *Oklahoma Academy of Science Technical Meeting* (November 2020, Virtual)